(P1) Given system of eq^n s, where $u(t)$ input & $y(t)$ output:

$$\frac{d}{dt}\begin{bmatrix} z_1(t) \\ z_2(t) \end{bmatrix} = \begin{bmatrix} z_1(t-1) + u(t) \\ z_1(t) \end{bmatrix},$$

$$y(t) = z_1(t+1) + z_2(t) - u(t)$$

N.B. non standard form of state space model & z_1, z_2 are not system's state variable.

The correct system characterizations are below.

① **Linear system**

There are nonlinear operations (e.g. z_i^2, $\sin(z_i)$, $z_i z_j$ ... etc) and $u(t)$, $z_1(t)$, $z_2(t)$ & $y(t)$ are in strictly linear relationship since every variable appear to the first power and variables are only added / subtracted.

$$\dot{z}_1(t) = z_1(t-1) + u(t) \Rightarrow z \text{ depend linearly on } u$$
$$\dot{z}_2(t) = z_1(t)$$
$$y(t) = z_1(t+1) + z_2(t) - u(t) \Rightarrow y(t) \text{ " " on } z \& u$$

so $y(t)$ depends on u & z and z depend on u and they are all linear dependence.

So if you scale input (or add input), the corresponding variables and outputs will scale and proportionally add, and will satisfy homogenity and additivity.

② **Continuous time**

This is easy because, system uses ODE's involving continuous time derivative $\left(\frac{d}{dt}\right)$ & in independent t varies continuously.

(3) **Time invariant**

The reason, all coefficients in system's differential and output variables in equations are constant. So all terms of z's and u are constant as a time shift $u(t \to t-\tau)$ will have identical time shift in the intermediate variables $z_1(t-\tau)$, $z_2(t-\tau)$ & output $y(t-\tau)$

So time invariant.

(4) **Non-Causal**

Since in output, $y(t) = z_1(t+1) + z_2(t) - u(t)$ at current t depend on $z_1(t+1)$ ie. one second in future. A causal system can depend only on past and present e.g. $z(t)$, $z(t-1)$... etc. But here $z_1(t+1)$ is future value. hence Non-Causal.

(5) **Distributed**

Here derivative $\dot{z}_1(t)$ depend on time delayed version of itself ie. $\dot{z}(t) = z_1(t-1) + u(t)$. For system to be lumped, it must be represented by finite number of state variables. Here knowing $z_1(t_0)$ and $z_2(t_0)$ at one instant is not enough to determine future behaviors. We need to know $z_1(\tau)$ over intervals like $[t_0-1, t_0]$. So the system needs infinitely many values to describe the state. Hence it's distributed.

As per lecture CT system and have time delays so distributed.

(P2) input $u \in \mathbb{R}$, state $x \in \mathbb{R}^2$, output $y \in \mathbb{R}$, $x(0)$ init state.

experiments

Exp① $x(0) = [1 \;\; 1]^T$ & $u(t) \equiv 1$ for $t \geq 0$, $y(t) = e^{-t}$, $t \geq 0$

Exp② $x(0) = [1 \;\; 1]^T$ & $u(t) \equiv -1$ for $t \geq 0$, $y(t) = e^{-2t}$, $t \geq 0$

Exp③ $x(0) = [1 \;\; -1]^T$ and $u(t) \equiv 1$ for $t \geq 0$, $y(t) = -e^{-t}$, $t \geq 0$.

'$\equiv$' means equals for all $t \geq 0$'

find $y(t)$, $t \geq 0$

a) $x(0) = [0 \;\; 0]^T$ and $u(t) \equiv 2$ for $t \geq 0$

Desired $x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $u(t) \equiv 2$

Say, exp① $x_1(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u \equiv 1$, $y_1(t) = e^{-t}$

exp② $x_2(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u \equiv -1$, $y_2(t) = e^{-2t}$

Subtract exp② from exp①,

$\therefore x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ & $u(t) = 1 - (-1) = 2$

$\therefore N(x(0), u(t) \equiv 2) = N\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, u(t) = 1 - (-1)\right)$

$N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, u(t) \equiv 2\right) = N(\underbrace{x(0), u(t) \equiv 1}_{\text{exp 1}}) - N(\underbrace{x(0), u(t) \equiv -1}_{\text{exp 2.}})$

Since the system is linear,

$y(t) = y_1(t) - y_2(t)$ at $t \geq 0$

$= e^{-t} - e^{-2t}$ Ans

b) To get $x(0) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ & $u(t) \equiv 0$ for $t \geq 0$,

we subtract exp③ from exp①

$\therefore x(0) = \begin{bmatrix} 1 \\ +1 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ & $u(t) = 1 - 1 = 0$

$\therefore$ system $\mathcal{N}\left(x(0), u \equiv 0\right) = \mathcal{N}\left(\begin{bmatrix}1\\1\end{bmatrix}, u(t) = 1\right) - \mathcal{N}\left(\begin{bmatrix}1\\-1\end{bmatrix}, u(t) = -1\right)$

$\mathcal{N}\left(\begin{bmatrix}0\\2\end{bmatrix}, u \equiv 0\right) = \underbrace{\mathcal{N}\left(\begin{bmatrix}1\\1\end{bmatrix}, u(t) \equiv 1\right)}_{\text{expt 1}} - \underbrace{\mathcal{N}\left(\begin{bmatrix}1\\-1\end{bmatrix}, u(t) \equiv 1\right)}_{\text{expt 3}}$

Since system linear $\therefore y(t) = y_1(t) - y_3(t)$

$= e^{-t} - (-e^{-t})$

$= 2e^{-t}, \quad t \geq 0$ Ans

(c) To get $x(0) = \begin{bmatrix}0\\1\end{bmatrix}$ & $u(t) \equiv -3$ for $t \geq 0$

Do let a, b, c are constants such that

$$a\begin{bmatrix}1\\1\end{bmatrix} + b\begin{bmatrix}1\\1\end{bmatrix} + c\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

& $a + b + c = -3.$

$\therefore$ $a + b + c = 0 \rightarrow c = -1/2 \Rightarrow a + b = 1/2$

$a + b - c = 1$

$a - b + c = -3.$ [for expt ② $u \equiv -1$, expt ① & ③ $u \equiv 1$]

$a - b - 1/2 = -3$

$a - b = -5/2$

$\therefore a + b = 1/2$

$\therefore a = -1, \quad b = 3/2$ & $c = -1/2$

$\mathcal{N}\left[x(0), u \equiv -3\right] = -\mathcal{N}\left(\begin{bmatrix}1\\1\end{bmatrix}, u \equiv 1\right) + \frac{3}{2}\mathcal{N}\left(\begin{bmatrix}1\\1\end{bmatrix}, u \equiv -1\right) - \frac{1}{2}\mathcal{N}\left(\begin{bmatrix}1\\-1\end{bmatrix}, u \equiv 1\right)$

$\therefore$ $\mathcal{N}\left(\begin{bmatrix}0\\1\end{bmatrix}, -3\right)$ =

due to linearity output

$$y(t) = -e^{-t} + \frac{3}{2}e^{-2t} - \frac{1}{2}e^{-t}$$

$$= -\frac{1}{2}e^{-t} + \frac{3}{2}e^{-2t}, \quad t \geq 0$$ Ans

Q.2.

(c) From part (a) & (b) we can also use existing result to come to (c).

from part (a) we get $\mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, 2\right) = e^{-t} - e^{-2t}$

(b) " " $\mathcal{N}\left(\begin{bmatrix}0\\2\end{bmatrix}, 0\right) = 2e^{-t}$ $\Big\}\ t \geq 0$

$\therefore$ Since $\left(\begin{bmatrix}0\\1\end{bmatrix}, -3\right) = \frac{1}{2}\left(\begin{bmatrix}0\\2\end{bmatrix}, 0\right) - 3/2\left(\begin{bmatrix}0\\0\end{bmatrix}, 2\right)$

or $\mathcal{N}\left(\begin{bmatrix}0\\1\end{bmatrix}, -3\right) = \frac{1}{2}\mathcal{N}\left(\begin{bmatrix}0\\2\end{bmatrix}, 0\right) - \frac{3}{2}\mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, 2\right)$

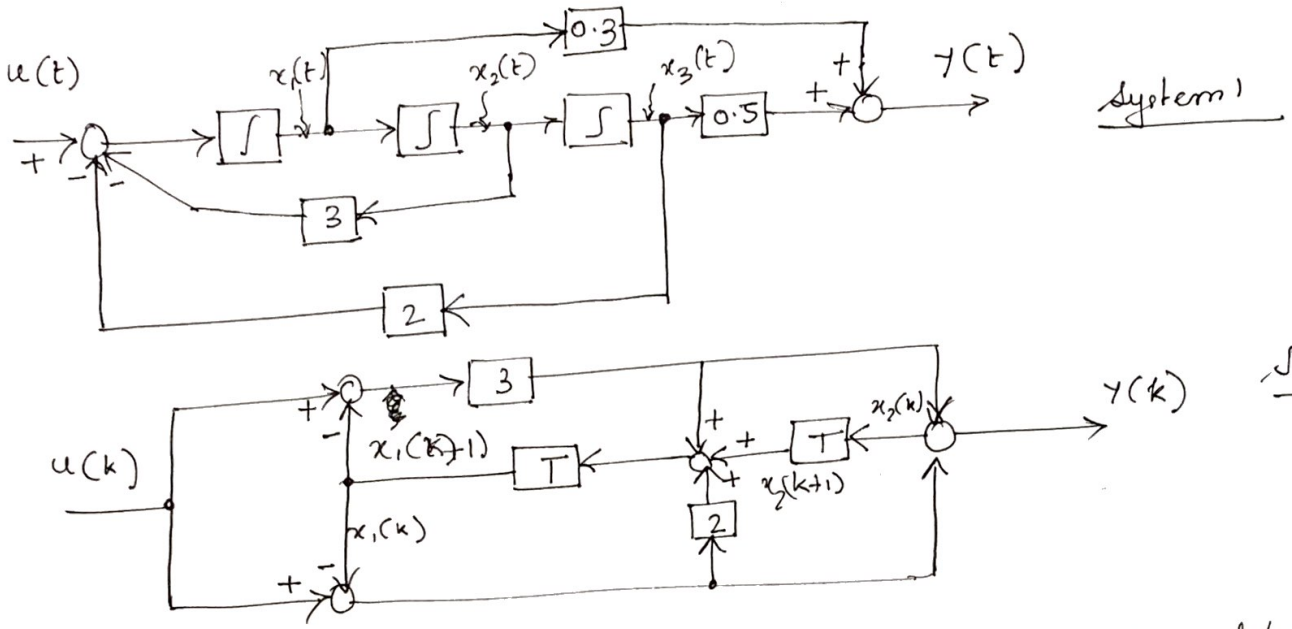
$\therefore$ using linearity,

$$y(t) = \frac{1}{2}\cdot(2e^{-t}) - 3/2\,(e^{-t} - e^{-2t})$$

$$= -\frac{1}{2}e^{-t} + 3/2\,e^{-2t}, \quad t \geq 0$$

Ans

If this method is unsatisfactory. I attempted a different method in page 16.



(P3) System 1 has 3 integrators so let, $x_1(t)$ be output of first integrator, $x_2(t)$ be for second & $x_3(t)$ be output of third integrator (from left to right)

State vector $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$

For x_1: input $\dot{x}_1(t) = u(t) - 3x_2(t) - 2x_3(t)$

For x_2: " $\dot{x}_2(t) = x_1(t)$

For x_3: " $\dot{x}_3(t) = x_2(t)$

$$\therefore \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & -3 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u$$

$$\therefore \dot{x} = A\,x + B.u$$

The output eqn $y(t) = 0.3x_1(t) + 0.5x_3(t)$

$$= \begin{bmatrix} 0.3 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + [0]u$$

$$y = C\,x + D.u$$

$$\therefore A = \begin{bmatrix} 0 & -3 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad C = [0.3, 0 \quad 0.5)$$

$$D = [0]$$

Ans

System ① Transfer function Model.

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D \quad \text{assuming zero initial cond.}$$

$$sI - A = \begin{bmatrix} s & +3 & +2 \\ -1 & s & 0 \\ 0 & -1 & s \end{bmatrix}$$

Let $(sI - A)\,v = B$ where $v = (sI - A)^{-1}B$

$$\therefore \begin{bmatrix} s & 3 & 2 \\ -1 & s & 0 \\ 0 & -1 & s \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$-v_2 + sv_3 = 0 \Rightarrow v_2 = sv_3$$

$$-v_1 + sv_2 = 0 \Rightarrow v_1 = sv_2 = s^2v_3$$

$$sv_1 + 3v_2 + 2v_3 = 1 \Rightarrow s^3v_3 + 3sv_3 + 2v_3 = 1.$$

$$\therefore v_3 = \frac{1}{s^3 + 3s + 2}, \quad v_1 = \frac{s^2}{s^3 + 3s + 2}$$

$$\& \quad v_2 = \frac{s}{s^3 + 3s + 2}$$

$$\therefore (sI - A)^{-1}B = \begin{bmatrix} \frac{s^2}{s^3+3s+2} \\ \frac{s}{s^3+3s+2} \\ \frac{1}{s^3+3s+2} \end{bmatrix}$$

$$\therefore \frac{Y(s)}{U(s)} = \underbrace{\begin{bmatrix} 0.3 & 0 & 0.5 \end{bmatrix}}_{C} \underbrace{\begin{bmatrix} \frac{s^2}{s^3+3s+2} \\ \frac{s}{s^3+3s+2} \\ \frac{1}{s^3+3s+2} \end{bmatrix}}_{(sI - A^{-1})B} + \underbrace{[0]}_{D}$$

$$H(s) \quad \frac{Y(s)}{U(s)} = \frac{0.3s^2}{s^3+3s+2} + \frac{0.5}{s^3+3s+2} = \frac{0.3s^2 + 0.5}{s^3+3s+2}$$

Ans

System 2 This is a discrete time system

$x_1(k) \rightarrow$ left time delay block

$x_2(k) \rightarrow$ right " " "

$$x[k] = \begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix}$$

Signal node:

Top left sum junction $\rightarrow u[k] - x_1[k]$

Bottom left " $\rightarrow u[k] - x_1[k]$

Output side:

Top branch $\rightarrow 3(u[k] - x_1[k])$

Bottom " $\rightarrow -[u[k] - x_1[k])$

summation junction

$$\therefore \quad y[k] = 3(u[k] - x_1[k]) - [u[k] - x_1[k])$$

$$v[k] = 2(u[k] - x_1[k]) \quad \text{— (1)}$$

Next, for state $k+1$,

$$x_2[k+1] = y[k]$$

$$x_2[k+1] = 2[u[k] - x_1[k]) \quad \text{from (1)}$$

$$x_2[k+1] = -2x_1[k] + 2u[k] \quad \text{— (2)}$$

middle junction three + inputs

$$3(u[k] - x_1[k]), 2(u[k] - x_1[k]), x_2[k]$$

$$x_1[k+1] = 3(u[k] - x_1[k]) + 2(u[k] - x_1[k]) + x_2[k]$$

$$x_1[k+1] = -5x_1[k] + x_2[k] + 5u[k] \quad \text{— (3)}$$

$$x_1[k+1] = -5x_1[k] + x_2[k] + 5u[k]$$

$$x_2[k+1] = -2x_1[k] + 0 \cdot x_2[k] + 2u[k]$$

$$y[k] = -2x_1[k] + 2u[k]$$

$$A = \begin{bmatrix} -5 & 1 \\ -2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \quad C = \begin{bmatrix} -2 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 2 \end{bmatrix}$$

Ans

Trans fer fenction calculations.

DT so z transform

$$H(z) = C(zI - A)^{-1}B + D.$$

$$(zI - A) = \begin{bmatrix} z+5 & -1 \\ +2 & z \end{bmatrix}$$

2 x 2 matrix $\therefore$ $\det(zI - A) = z(z+5) + 2$

$$= z^2 + 5z + 2$$

$\therefore$ determinant $= z^2 + 5z + 2$

$$\& \quad (zI - A)^{-1} = \frac{1}{z^2+5z+2} \begin{bmatrix} z & 1 \\ -2 & z+5 \end{bmatrix}$$

$$\therefore \quad (zI - A)^{-1}B = \frac{1}{z^2+5z+2} \begin{bmatrix} z & 1 \\ -2 & z+5 \end{bmatrix}_{2\times 2} \begin{bmatrix} 5 \\ 2 \end{bmatrix}_{2\times 1}$$

$$= \frac{1}{z^2+5z+2} \begin{bmatrix} 5z+2 \\ 2z \end{bmatrix}_{2\times 1}$$

$$C(zI - A)^{-1}B = \begin{bmatrix} -2 & 0 \end{bmatrix}_{1\times 2} \frac{1}{z^2+5z+2} \begin{bmatrix} 5z+2 \\ 2z \end{bmatrix}$$

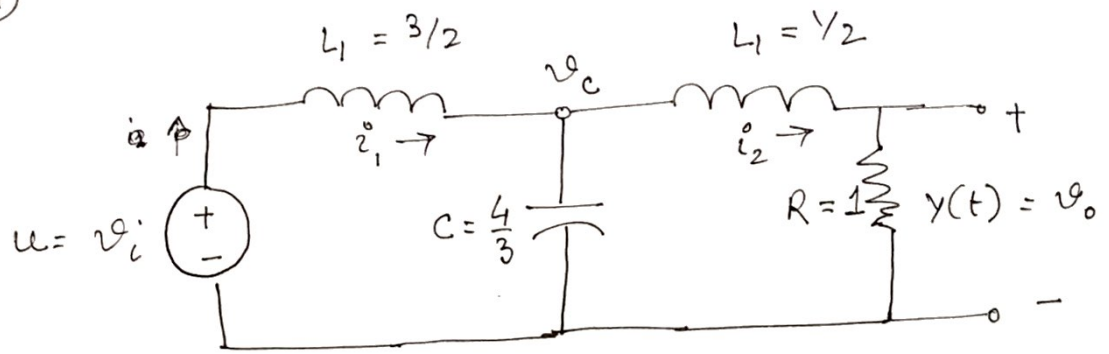
$$= \frac{1}{z^2+5z+2} \begin{bmatrix} -10z - 4 \end{bmatrix}$$

$$C(zI - A)^{-1}B + D = \frac{1}{z^2+5z+2}(-10z - 4) + 2$$

$$= \frac{-10z - 4 + 2z^2 + 10z + 4}{z^2+5z+2}$$

$$= \frac{2z^2}{z^2+5z+2} \quad \text{Ans} \quad \left| \frac{\text{or}}{1+5z^{-1}+2z^{-2}} \right. \quad \frac{2}{}$$

(P4)



Let state variables are,

$x_1 = i_1$, $x_2 = v_c$, $x_3 = i_2$ | $u = v_i$, $y(t) = v_o$

eqn for $x_1 = \dot{i}_1$

$v_{Li} = L_1 \frac{di_1}{dt}$, here $L_1 = 3/2$ & $v_{Li} = v_i - v_c$

$\therefore$ $\therefore \frac{3}{2} \frac{di_1}{dt} = v_i - v_c$

$\frac{3}{2} \dot{x}_1 = u - x_2$

or $\dot{x}_1 = \frac{2}{3} u - \frac{2}{3} x_2$ —①

eqn for $x_2 = v_c$ $i_c = C \frac{dv_c}{dt}$ here $C = \frac{4}{3}$ $\therefore i_c = \frac{4}{3} \frac{dv_c}{dt}$

$\therefore$ $i_1 = i_c + i_2$ kirchoff?

$i_c = i_2 - i_1$

$\frac{4}{3} \frac{dv_c}{dt} = i_2 - i_1$ or, $\frac{4}{3} \dot{x} = x_3 - x_1$

or $\dot{x} = \frac{3}{4} x_3 - \frac{3}{4} x_1$ —②

eqn for $x_3 = \dot{i}_2$

$v_{L_2} = L_2 \frac{di_2}{dt}$, $L_2 = 1/2$

$\therefore \frac{1}{2} \frac{di_2}{dt} = v_c - v_o$ or $\frac{1}{2} \dot{x}_3 = x_2 - v_o$

$= x_2 - R i_2$

$= x_2 - i_2$

or $\frac{1}{2}\dot{x}_3 = x_2 - i_2 = x_2 - x_3$ since $i_2 = x_3$ & $R = 1$.

$\therefore \quad \dot{x}_3 = 2x_2 - 2x_3$ —③

$\therefore$ eqn ①, ② & ③ are

$$\dot{x}_1 = 0\,x_1 - \frac{2}{3}x_2 + \frac{2}{3}u + 0 \cdot x_3.$$

$$\dot{x}_2 = \frac{3}{4}x_1 + 0 \cdot x_2 - \frac{3}{4}x_3$$

$$\dot{x}_3 = 0\,x_1 + 2x_2 - 2x_3$$

Std form $\dot{x} = Ax + Bu$

$$\therefore \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -2/3 & 0 \\ 3/4 & 0 & -3/4 \\ 0 & 2 & -2 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 2/3 \\ 0 \\ 0 \end{bmatrix}}_{B} u \quad \text{—④}$$

Output $y = v_o = R i_2 = i_2 = x_3$ or $y = x_3$

Std form $y = Cx + Du$

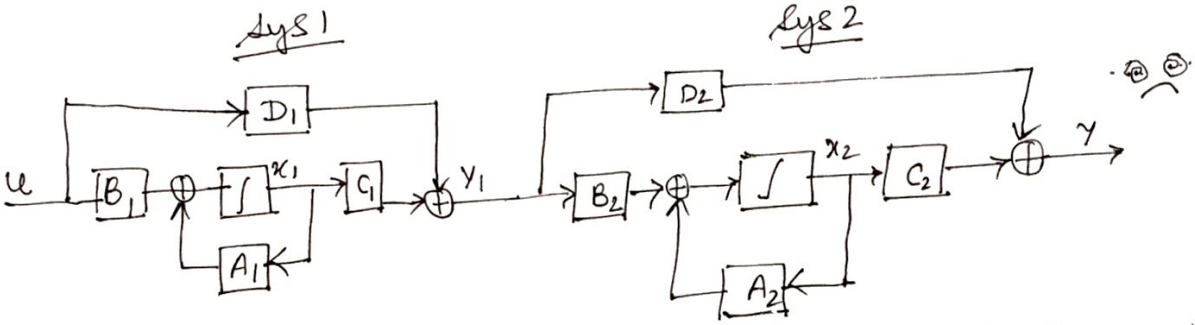
$$= \underbrace{\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}}_{C} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{0}_{D} \cdot u \quad \text{—⑤}$$

$$A = \begin{bmatrix} 0 & -2/3 & 0 \\ 3/4 & 0 & -3/4 \\ 0 & 2 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 2/3 \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \quad D = [0]$$

with $x_1 = i_1$, $x_2 = v_c$ & $x_3 = i_2$

Answer

① Serial Connection



$H_{series} = H_2(s)\, H_1(s)$

Find A, B, C, D in terms of subsystems!

Sys 1

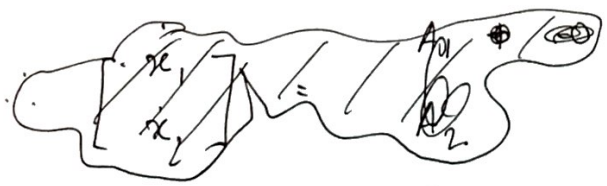
$$\dot{x}_1(t) = A_1 x_1(t) + B_1 u(t), \quad \dot{x}_1 = A_1 x_1 + B_1 u \quad \text{—— (1)}$$

$$y_1(t) = C_1 x_1(t) + D_1 u(t), \quad y_1 = C_1 x_1 + D_1 u \quad \text{—— (2)}$$

Sys 2

$$\dot{x}_2(t) = A_2 x_2(t) + B_2 y_1(t), \quad \dot{x}_2 = A_2 x_2 + B_2 y_1 \quad \text{—— (3)}$$

$$y_2(t) = C_2 x_2(t) + D_2 y_1(t), \quad y_2 = C_2 x_2 + D_2 y_1 \quad \text{—— (4)}$$



y_1 from (2) in (3),

$$\dot{x}_2 = A_2 x_2 + B_2 C_1 x_1 + B_2 D_1 u \quad \text{—— (5)}$$

∴ from (1) & (5)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 x_1 + B_1 u \\ B_2 C_1 x_1 + A_2 x_2 + B_2 D_1 u \end{bmatrix} \quad \text{—— (6)}$$

$$= \underbrace{\begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} B_1 \\ B_2 D_1 \end{bmatrix}}_{B} u$$

$$y_2 = C_2 x_2 + D_2 y_1 \quad \text{—④}$$

Substituting y_1 from eqn ② into ④

$$y = C_2 x_2 + D_2 (C_1 x_1 + D_1 u)$$

$$= D_2 C_1 x_1 + C_2 x_2 + D_2 D_1 u$$

~~$y = Cx + Du$~~

$$\text{or } y = \underbrace{[D_2 C_1 \quad C_2]}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{D_2 D_1}_{D} u \quad \text{—⑦}$$

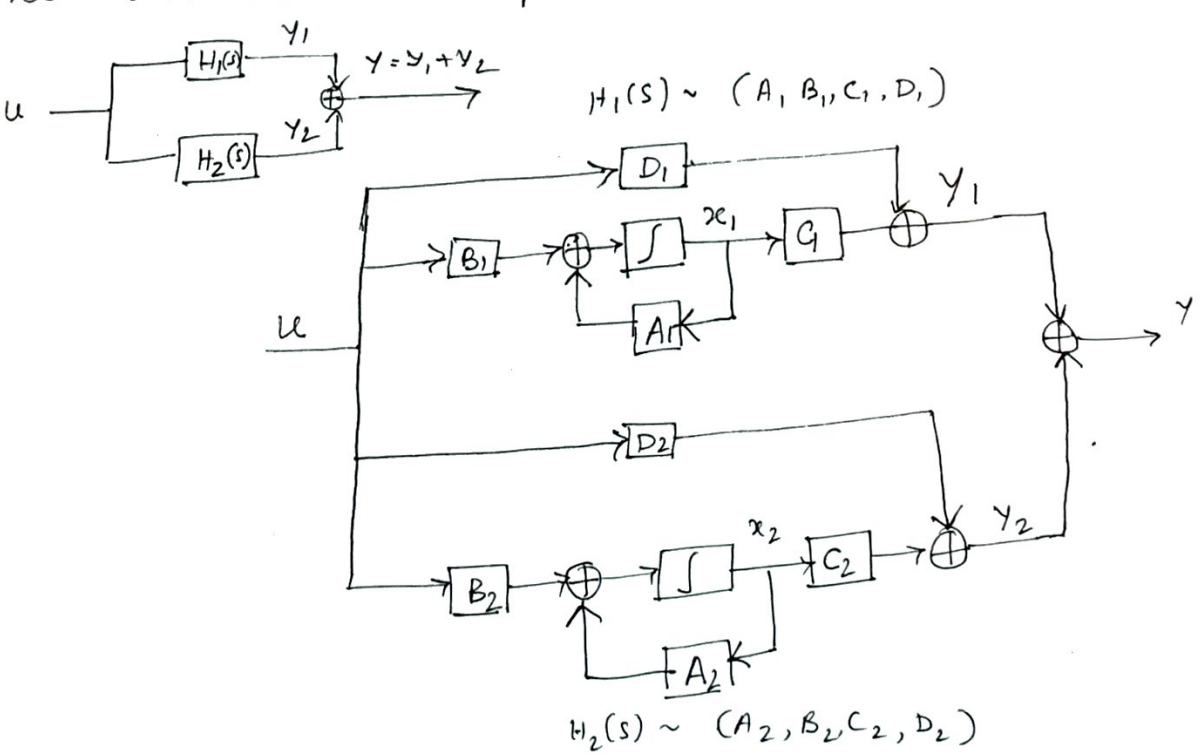
$$\therefore A = \begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 D_1 \end{bmatrix}, \quad C = [D_2 C_1 \quad C_2], \quad D = [D_2 D_1]$$

$$\dot{x} = \begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 D_1 \end{bmatrix} u$$

$$y = [D_2 C_1 \quad C_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + D_2 D_1 u. \quad \text{Answer}$$

P5 ②

Parallel connection: $H_{parallel} = H_1(s) + H_2(s)$



$\dot{x}_1 = A_1 x_1 + B_1 u \quad , \quad y_1 = C_1 x_1 + D_1 u$ Sys ①

& $\dot{x}_2 = A_2 x_2 + B_2 u \quad , \quad y_2 = C_2 x_2 + D_2 u$. Sys ②

Same input u & outputs added for parallel so $y = y_1 + y_2$

$$\therefore \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 x_1 + B_1 u \\ A_2 x_2 + B_2 u \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u \quad \text{— ①}$$

$$\therefore A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

$$y = (C_1 x_1 + D_1 u) + (C_2 x_2 + D_2 u)$$

$$= C_1 x_1 + C_2 x_2 + (D_1 + D_2) u$$

$$y = \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + (D_1 + D_2)[u] \quad \text{— ②}$$

$$\therefore C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}, \quad D = (D_1 + D_2)$$

& state space eqn are realized using eqn ① & ②

where, $A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}, \quad D = D_1 + D_2$

Answer

P5

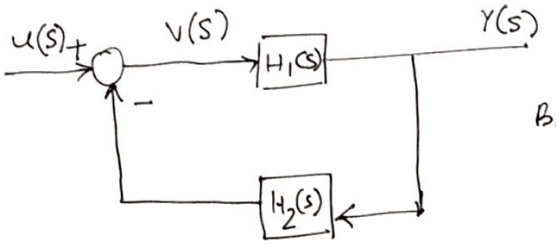
③ Negative feedback connection

$H_1(s)$ feed forward path

$H_2(s)$ negative feed forward path.

So, $H_{feedback} = (I + H_1(s) H_2(s))^{-1} H_1(s)$.

from page 113 4th ed Math systems theory [Olsder / Jetsema et al]



$V(s) = U(s) - H_2(s)Y(s)$

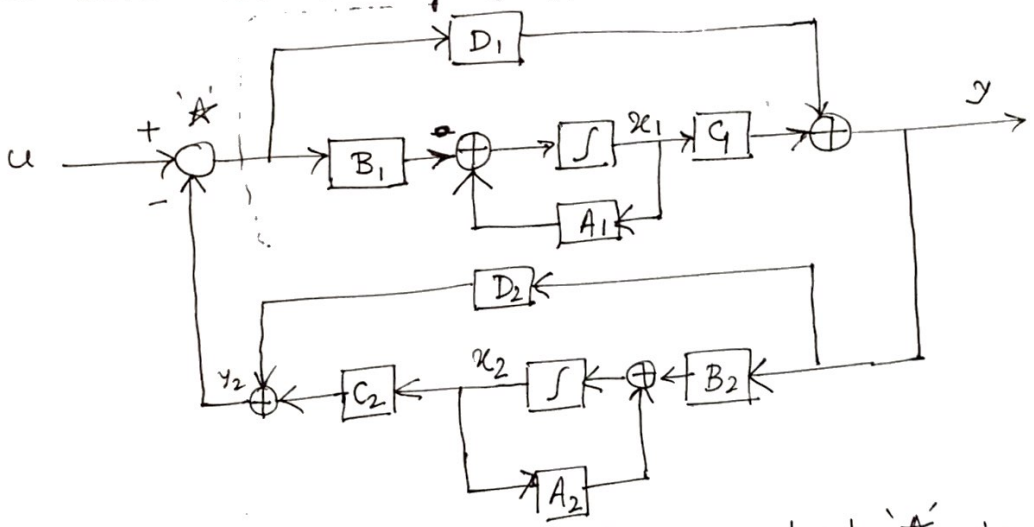
$Y(s) = H_1(s)V(s)$

Broadly $Y(s) = H_1(s)(U(s) - H_2(s)Y(s))$

or $Y(s) = (1 + H_1(s)H_2(s))^{-1} H_1(s)$

$\frac{Y(s)}{U(s)} = H(s) = (1 + H_1(s)H_2(s))^{-1} H_1(s)$

We can construct our system as, ...



for H_1 signal at summing junction marked 'A' is $u - (C_2 x_2 + D_2 y)$

$$\dot{x}_1 = A_1 x_1 + B_1 (u - C_2 x_2 - D_2 y) \quad \text{---(1)}$$

$$y = C_1 x_1 + D_1 (u - C_2 x_2 - D_2 y) \quad ?? \quad \text{---(2)}$$

$$= C_1 x_1 + D_1 u - D_1 C_2 x_2 - D_1 D_2 y$$

and for H_2, $\dot{x}_2 = A_2 x_2 + B_2 y$ ---(3)

$y_{e2} =$ [illegible]

$y_2 = C_2 x_2 + D_2 y$

from (2)

$$y = C_1 x_1 + D_1 u - D_1 C_2 x_2 - D_1 D_2 y$$

$$y + D_1 D_2 y = C_1 x_1 - D_1 C_2 x_2 + D_1 u$$

$$(I + D_1 D_2) y = C_1 x_1 - D_1 C_2 x_2 + D_1 u$$

let $(I + D_1 D_2)^{-1} = E$

crazy! emm... should I assume $I + D_1 D_2$ is invertible? ok

$$y = E C_1 x_1 - E D_1 C_2 x_2 + E D_1 u \quad \text{—— (4)}$$

$$\therefore \quad y = \begin{bmatrix} E C_1 & -E D_1 C_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + E D_1 u$$

$$\therefore \quad C = \begin{bmatrix} E C_1 & - E D_1 C_2 \end{bmatrix}$$

$$D = E D_1$$

from ① $\quad \dot{x}_1 = A_1 x_1 + B_1 (u - C_2 x_2 - D_2 y)$

or $\dot{x}_1 = A_1 x_1 + B_1 u - B_1 C_2 x_2 - B_1 D_2 (E C_1 x_1 - E D_1 C_2 x_2 + E D_1 u)$

Substituted y from (4)

$$\dot{x}_1 = (A_1 - B_1 D_2 E C_1) x_1 + (- B_1 C_2 + B_1 D_2 E D_1 C_2) x_2 + (B_1 - B_1 D_2 E D_1) u. \quad \text{—— (5)}$$

from ③, $\quad \dot{x}_2 = A_2 x_2 + B_2 y$

$= A_2 x_2 + B_2 (E C_1 x_1 - E D_1 C_2 x_2 + E D_1 u)$

Substituted y from 4 for y,

$$\dot{x}_2 = B_2 E C_1 x_1 + (A_2 - B_2 E D_1 C_2) x_2 + B_2 E D_1 u \quad \text{—— (6)}$$

Model eqn : $\dot{x} = Ax + Bu \quad | \quad y = Cx + Du$

$$\dot{x}_1 = (A_1 - B_1 D_2 E C_1) x_1 + (- B_1 C_2 + B_1 D_2 E D_1 C_2) x_2 + (B_1 - B_1 D_2 E D_1) u$$

$$\dot{x}_2 = B_2 E C_1 x_1 + (A_2 - B_2 E D_1 C_2) x_2 + B_2 E D_1 u.$$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 - B_1 D_2 E C_1 & - B_1 C_2 + B_1 D_2 E D_1 C_2 \\ B_2 E C_1 & A_2 - B_2 E D_1 C_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B_1 - B_1 D_2 E D_1 \\ B_2 E D_1 \end{bmatrix} u \quad \text{—— (7)}$$

$$\therefore A = \begin{bmatrix} A_1 - B_1 D_2 E C_1 & -B_1 C_2 + B_1 D_2 E D_1 C_2 \\ B_2 E C_1 & A_2 - B_2 E D_1 C_2 \end{bmatrix}$$

$$B = \begin{bmatrix} B_1 - B_1 D_2 E D_1 \\ B_2 E D_1 \end{bmatrix}$$

$$y = \begin{bmatrix} E C_1 & -E D_1 C_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + E D_1 u \quad \text{—— (8)}$$

so $C = \begin{bmatrix} EC & -E D_1 C_2 \end{bmatrix}$, $D = E D_1$ & $E = (I + D_1 D_2)^{-1}$

Prob 2 [trying another method here]

(a) The response of linear system N can be decomposed as,

$$N(u|_{t \geq 0}, x(t_0)) = N(u|_{t \geq t_0}, 0) + N(0, x(t_0))$$

zero-state response (zsr); zero-input response (zir)

exp ① $x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u(t) \equiv 1$, $t \geq 0$

$$\therefore y_{zsr}(1) + y_{zir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = e^{-t} \quad \text{—— ①}$$

exp ② $x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u(t) \equiv -1$, $t \geq 0$

$$\therefore y_{zsr}(-1) + y_{zir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = e^{-2t} \quad \text{or} \quad -y_{zsr}(1) + y_{zir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = e^{-2t} \quad \text{—— ②}$$

(↑ homogeneity)

exp (3) $x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $u(t) \equiv 1$, $t \geq 0$

$$\therefore y_{zsr}(1) + y_{zir}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = -e^{-t} \quad \text{—— ③}$$

from ① − ② $\quad 2 y_{zsr}(1) = e^{-t} - e^{-2t} \quad y_{zsr}(1) = \frac{1}{2} e^{-t} - \frac{1}{2} e^{-2t}$

from ① + ② $\quad 2 y_{zir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = e^{-t} + e^{-2t} \quad y_{zir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \frac{1}{2} e^{-t} + \frac{1}{2} e^{-2t}$

from ③ add y_{zsr} value, $\left(\frac{1}{2} e^{-t} - \frac{1}{2} e^{-2t}\right) + y_{zir}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = -e^{-t}$

$$y_{zir}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = -\frac{3}{2} e^{-t} + \frac{1}{2} e^{-2t}$$

Case (a) $x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $u(t) \equiv 2$ for $t \geq 0$,

$y_{zir}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = 0$ & $y_{zsr}(2) = 2 y_{zsr}(1)$

$$\therefore y(t) = y_{zsr}(2) + y_{zir}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)$$

$$y(t) = 2\left(\frac{1}{2}e^{-t} - \frac{1}{2}e^{-2t}\right) = \underline{e^{-t} - e^{-2t}} \quad \text{Ans.}$$

Case (b) $x(0) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, & $u(t) \equiv 0$, $t \geq 0$

$y_{3ir}\left(\begin{bmatrix} 0 \\ 2 \end{bmatrix}\right)$ & $y_{3sr} = 0$ since $u = 0$

$$\therefore y(t) = y_{3sr}(0) + y_{3ir}\left(\begin{bmatrix} 0 \\ 2 \end{bmatrix}\right)$$

$$= y_{3sr}(0) + y_{3ir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) - y_{3ir}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$$

$$= \frac{1}{2}e^{-t} + \frac{1}{2}e^{-2t} + \frac{3}{2}e^{-t} - \frac{1}{2}e^{-2t}$$

$$y(t) = \underline{2e^{-t}} \quad \text{Answer}$$

Case (c)

$x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $u(t) \equiv -3$, $t \geq 0$

$y_{3ir}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$ & $y_{3sr} = -3$

$$y(t) = y_{3sr}(-3) + y_{3ir}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

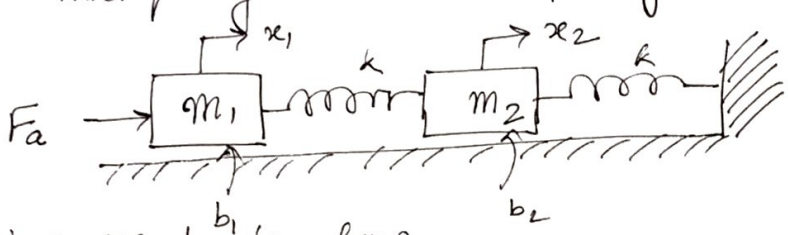
$$= -3 \cdot y_{3sr}(1) + \frac{1}{2}\left[y_{3ir}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) - y_{3ir}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)\right]$$

$$= -3\left(\frac{1}{2}e^{-t} - \frac{1}{2}e^{-2t}\right) + \frac{1}{2}\left[\frac{1}{2}e^{-t} + \frac{1}{2}e^{-2t} + \frac{3}{2}e^{-t} - \frac{1}{2}e^{-2t}\right]$$

$$= -\frac{3}{2}e^{-t} + \frac{3}{2}e^{-2t} + e^{-t}$$

$$= \underline{-\frac{1}{2}e^{-t} + \frac{3}{2}e^{-2t}} \quad \underline{\text{Ans}}$$

P4 (2) Attempting mass and spring



$u = F_a$

$y = \dot{z}_2$

$z_1, z_2 \rightarrow$ displacements

$F = m \cdot \ddot{z}$ ~~Hooke law~~

Using newton's law,

$$m_1 \ddot{z}_1 = F_a - b_1 \dot{z}_1 + k(z_2 - z_1)$$

$$= u - b_1 \dot{z}_1 + k z_2 - k z_1$$

$$\ddot{z}_1 = \frac{u}{m_1} - \frac{b_1}{m_1}\dot{z}_1 + \frac{k}{m_1} z_2 - \frac{k}{m_1} z_1 \quad \text{——} (1)$$

$$m_2 \ddot{z}_2 = k(z_1 - z_2) - k z_2 - b \dot{z}_2$$

$$= k z_1 - k z_2 - k z_2 - b \dot{z}_2$$

$$\ddot{z}_2 = \frac{k}{m_2} z_1 - \frac{2k}{m_2} z_2 - \frac{b}{m_2} \dot{z}_2 \quad \text{——} (2)$$

$$x(t) = \begin{bmatrix} z_1(t) \\ \dot{z}_1(t) \\ z_2(t) \\ \dot{z}_2(t) \end{bmatrix}$$ positions & velocities of masses 1 & 2

CT state space eq^n

$$\dot{x} = A x(t) + B u(t) \quad \text{——} (3)$$

$$y(t) = C x(t) + D u(t) \quad \text{——} (4)$$

~~$x_1 = z_1$, $x_2 = \dot{z}_1$~~

$\therefore x_1 = z_1$, $\underline{\dot{x}_1 = \dot{z}_1 = x_2}$

also $x_2 = \dot{z}_1$

$\dot{x}_2 = \ddot{z}_1$

from (1) $\dot{x}_2 = \frac{u}{m_1} - \frac{b_1}{m_1}\dot{z}_1 + \frac{k}{m_1} z_2 - \frac{k}{m_1} z_1 \quad \text{——} (5)$

$x_3 = z_2$ so $\underline{\dot{x}_3 = \dot{z}_2 = x_4}$ since (3) $x_4 = \dot{z}_2$

$x_4 = \dot{z}_2$ so, $\dot{x}_4 = \ddot{z}_2 = \frac{k}{m_2} z_1 - \frac{2k}{m_2} z_2 - \frac{b}{m_2} \dot{z}_2 \quad \text{——} (6)$

(5) & (6) replacing with state variables

(5) become $\underline{\dot{x}_2} = -\frac{k}{m_1} x_1 - \frac{b_1}{m_1} x_2 + \frac{k}{m_1} x_3 + \frac{1}{m_1} u$

(6) becomes $\underline{\dot{x}_4} = \frac{k}{m_2} x_1 - \frac{2k}{m_2} x_3 - \frac{b}{m_2} x_4$

State space eqn ③

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -k/m_1 & -b/m_1 & k/m_1 & 0 \\ 0 & 0 & 0 & 1 \\ k/m_2 & 0 & -2k/m_2 & -b/m_2 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1/m_1 \\ 0 \\ 0 \end{bmatrix}}_{B} u \quad \text{—①}$$

$y = \dot{x}_2 = x_4$

$\therefore$ eqn ② $\Rightarrow y = \underbrace{[0\ 0\ 0\ 1]}_{C} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \underbrace{0}_{D} . u$

Answer.